XIAOTIAN CHEN

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EDUCATION

Columbia University New York, NY
M.S. in Electrical Engineering Expected Dec 2027

The University of Nottingham Ningbo China Ningbo, China
Bachelor of Engineering in Electrical and Electronic Engineering Graduation Date: Jul 2026
GPA: 3.5/4.0

WORK EXPERIENCE

Ningbo Ausstattung Machinery Technology Co., Ltd. | R&D Department Ningbo, China
Algorithm Engineer Intern Jun 2026 - Aug 2026

- Architected the electrical and control system of a five-actuator inspection robot, integrating motor drives, CAN/CAN-FD communication, power distribution, and operator interfaces.
- Developed multi-joint motion control logic with synchronized actuation, joint-specific velocity profiles, position limits, homing, and emergency-stop handling.
- Built a Python control platform with real-time motor feedback, 20 Hz trajectory recording, keyframe editing, interpolation, and adjustable replay speeds (0.5–2×).
- Led system integration and debugging across motor firmware, CAN communication, feedback latency, and cable-driven mechanisms to improve motion reliability.

SELECTED ROBOTICS & AI PROJECTS

Indoor Inspection Robot - Autonomous Navigation System | Researcher Jul 2024 - Sep 2024

- Developed a ROS 2-based autonomous navigation stack, integrating LiDAR, IMU, and wheel odometry for SLAM and multi-sensor localization, reducing localization error from 0.30 m to 0.05 m.
- Implemented real-time obstacle detection and avoidance using 10 Hz LiDAR data, achieving sub-0.5 s response times and enabling four hours of collision-free indoor navigation.
- Engineered ROS 2 monitoring nodes in Python for sensor diagnostics and battery-state tracking, achieving approximately 95% alert precision during extended autonomous operation.

Development of UAV Autonomous Target Search & Navigation System | Research Assistant May 2024 - Jul 2024

- Developed a ROS 2–PX4 autonomous navigation framework on Ubuntu 22.04, integrating ROS 2 Humble, Gazebo simulation, and modular publish–subscribe communication for robotic control.
- Implemented and optimized motion planning and control algorithms, including A*, RRT*, PID, and MPC, achieving trajectory-tracking errors within ±0.2 m (square) and ±0.1 m (circular), with velocity fluctuations below 5%.
- Integrated visual SLAM, YOLOv8-based target detection, and EKF-based multi-sensor fusion to support autonomous target search, localization, and trajectory execution.

Multi-Static Passive Wi-Fi Human Activity Recognition with CSI | Independent Researcher Aug 2025 - Jun 2026

- Developed a distributed Wi-Fi CSI sensing system using three Raspberry Pi nodes and Nexmon, with synchronized data acquisition and signal preprocessing.
- Trained PyTorch activity-recognition models and evaluated multi-sensor fusion, weighted decision fusion, and cross-scenario generalization.

COMPETITION

Intelligent Robot Design Competition of UNNC | Project Leader Oct 2023 - Jan 2024

- Led a team to build an Arduino-based autonomous mobile robot, winning 1st Prize in the UNNC Robotics Competition.
- Implemented encoder-feedback PID motor control with PWM actuation and Ziegler–Nichols tuning, achieving ±2 cm trajectory deviation and less than 3% speed variation during five-minute figure-eight navigation.

TECHNICAL SKILLS

Languages: Python, C++, C, MATLAB
Robotics & Embedded: ROS 2, SLAM, PX4, PID, Sensor Fusion, STM32, CAN/CAN-FD
Tools: Linux, Gazebo, PyTorch, MATLAB/Simulink, Arduino